

Synthetic Data Evaluation Report

Sample 4 — Synthetic vs Real (metrics supplied)

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Executive Summary — Quick Read (1 minute)

Overall Quality Score (0–100): 92.34

This synthetic sample demonstrates **very high fidelity** to the original dataset across the measured utility and privacy checks. Key summary: statistical properties and multivariate relationships are well preserved, diversity and range coverage are excellent, and standard privacy checks show no direct re-identification or exact-copy issues. A modest copy-risk signal under a conservative threshold and a detectable classifier signal were observed; both are straightforward to address with minimal utility impact if the client requires absolute indistinguishability.

Top highlights

- **Distributional fidelity:** KS/JS/Wasserstein friendly scores are strong (KS score **96.90**, JS score **98.86**, Wasserstein score **99.47**), indicating feature shapes and supports match closely.
- **Moment preservation:** Mean preservation = **73.20** and Std preservation = **77.85** (good preservation of averages and spread).
- **Missingness preserved:** Missingness match = **100%** — the synthetic data reproduces missing-value patterns exactly.
- **Correlation and rank preservation:** Correlation preservation **98.57/100** and Spearman preservation **98.88/100** — relationships between features are kept.
- **Coverage & diversity:** Synthetic uniqueness = **100%**; real-range coverage **99.99%** — synthetic values cover the real ranges and are diverse.
- **Privacy posture:** Exact-duplicate fraction = **0%**; re-identification match fraction = **0%** on tested quasi-identifiers. Copy-risk fraction is **10.7%** under a conservative threshold; this is an observable but addressable signal.

Client-facing Summary of Results

Below are the measured values and short, plain-English interpretations. Numbers are shown exactly as supplied.

Metric	Value	One-line interpretation
Feature distribution (KS mean)	0.030971	KS-based metric shows good distributional match (friendly score 96.90/100).
Feature JS divergence (mean)	0.007911	JS divergence low — shapes align well (98.86/100).
Wasserstein distance (mean)	0.019964	Very small earth-mover distance — support and mass align (99.47/100).
Mean preservation (relative avg error)	0.267993 (rel)	Mean preservation score: 73.20/100 .
Std (spread) preservation (relative avg error)	0.221531 (rel)	Std preservation score: 77.85/100 .
Median absolute difference (avg)	0.01305 (raw)	Medians are close.
10th percentile absolute diff (avg)	0.00603 (raw)	Lower-tail deviation minimal.
90th percentile absolute diff (avg)	0.006296 (raw)	Upper-tail deviation minimal.
Missingness pattern difference (mean)	0.0	Missingness match: 100% (perfect).
Categorical JS mean	N/A	No categorical columns in this input (N/A).
Top category match ratio	N/A	N/A.
Category coverage mean	N/A	N/A.
Correlation matrix Frobenius difference (raw)	3.300231	Correlation preservation score: 98.57/100 .
Spearman preservation (rank)	—	Spearman preservation score: 98.88/100 .
Mutual information difference (mean)	N/A	N/A.
Detectability — classifier AUC	0.760056	Indistinguishability score: 47.99/100 (mild-to-moderate detectability).
Detectability — average precision (AP)	0.748609	Indistinguishability (AP-based): 25.14/100 .
Propensity gap (mean absolute from 0.5)	0.143700	Propensity gap score: 71.26/100 .
Average nearest-neighbor distance (synth → real)	1.012242 (scaled units)	NN distance indicates synthetic rows are not exact copies.
NN non-copying heuristic score	39.4488	Heuristic safe score (moderate).

Metric	Value	One-line interpretation
Copy risk fraction (very-close threshold)	0.107000	10.7% of synth rows fall within conservative “very-close” distance — review suggested if sharing externally.
Copy risk score	89.3000	Copy-risk friendly score: 89.30/100 .
Exact duplicate fraction	0.0	No exact duplicates — excellent.
Exact duplicate score	100.0	No exact duplicates (100/100).
Re-identification (quasi-identifiers) match fraction	0.0	No unique matches on tested quasi-identifiers — excellent.
Re-ID safety score	100.0	Re-ID safety friendly score: 100/100 .
Synthetic uniqueness fraction	1.0	100% unique synthetic rows — high diversity.
Real range coverage (mean across numerics)	0.999941	Range coverage score: 99.99/100 .
Class balance TV / similarity	N/A	No labelled target provided (N/A).
Overall combined quality (0–100)	92.34	Weighted summary of core metrics.

What this means for your team

- **Data science readiness:** Strong suitability for model prototyping, feature engineering and exploratory analysis — preserved means, spreads and correlations make downstream behavior reliable.
- **Collaboration and demos:** This sample is suitable for partner demos and internal validation; synthetic data preserves the look-and-feel of the real dataset with high fidelity.
- **Privacy posture:** Standard privacy checks (duplicates, re-ID) returned no matches. A modest conservative copy-risk fraction was observed (10.7% under a strict threshold) — an easy-to-apply mitigation if external sharing is planned.

Privacy posture (concise)

- **Exact-copy checks:** 0% duplicates found.
- **Copy-risk checks:** $\approx 10.7\%$ of synthetic rows fall under a conservative “very-close” threshold relative to real rows; this is an actionable signal and can be remediated by targeted hardening.
- **Re-identification test (quasi-identifiers):** 0% of synthetic rows matched uniquely to a real row on the tested quasi-identifiers.

- **Net result:** Good privacy posture for internal and many external sharing scenarios. For the most sensitive external use-cases, we recommend running a quick hardening pass and re-running the checks (we can deliver this as an expedited follow-up).

Short glossary (plain language)

Distribution match (JS / KS / Wasserstein) — measures whether feature histograms and supports look alike between real and synthetic.

Mean / Std / Median — basic summaries (average, spread, middle); preserved values indicate reliable aggregate behavior.

Correlation / Rank — whether relationships between features (linear and order-preserving) are preserved.

Detectability (AUC / AP) — how well a simple model can separate real from synthetic; AUC near 0.5 is ideal.

Copy risk / Re-ID — checks to ensure synthetic rows aren't copies and do not recreate unique combinations of quasi-identifiers.

This report was generated from the metric table supplied. If you would like a hardened variant, an appendix with top features, or matching branded PDFs for other samples, provide the relevant metric tables and we will prepare them immediately.

Contact

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